

D2 finance Audit Summary

Prepared by Cyfrin

Version 2.1

Lead Auditors

Immeas

February 24, 2025

Contents

1 About Cyfrin **2**

2 About This Document **2**

3 Disclaimer **2**

4 Copyright and License **2**

5 Audit Overview **3**

6 Audit Scope **3**

7 Risk Classification **4**

8 Protocol Summary **4**

8.1 Key Components 4

8.2 VaultV3 Flow 5

9 Attestation **5**

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for D2 on D2 finance. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated February 24, 2025, was delivered to D2, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from D2 or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 26 days, the Cyfrin team conducted an audit on the D2 finance code provided by D2. The findings consist of 2 Critical, 2 High, 7 Medium & 7 Low severity issues with the remainder being informational.

Summary

Project Name	D2 finance
Repository	d2-contracts
Commit	c2fc257605eb...
Audit Timeline	Jan 10th - Feb 14th, 2025
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	2	2	0
High Risk	2	2	0
Medium Risk	7	7	0
Low Risk	7	4	3
Informational	6	5	1
Gas Optimizations	0	0	0
Total	24	20	4

6 Audit Scope

Cyfrin conducted an audit of D2 based on the code present in the repository commit hash `c2fc257`. Cyfrin also audited the additions of a Kodiak Island integration in commit `ca2e36e` as well as the additions in commit `ab5b852`.

The following contracts were included in the scope of the audit:

```
contracts/modules/Aave.sol
contracts/modules/Bera.sol
contracts/modules/Camelot.sol
contracts/modules/D2.sol
contracts/modules/Dolomite.sol
contracts/modules/GMXV2.sol
contracts/modules/IDolomite.sol
contracts/modules/Inch.sol
contracts/modules/Pendle.sol
contracts/modules/Silo.sol
contracts/modules/Trader.sol
contracts/modules/WETH.sol
contracts/D2OFT.sol
contracts/Strategy.sol
contracts/VaultV0.sol
contracts/VaultV3.sol
```

- `contracts/modules/Bera.sol` includes integrations with Bera staking, Kodiak, Ooga Booga, Infrared and Dolomite.
- `contracts/modules/GMXV2.sol` had the `GMXV2_Module` contract commented out; it was treated as in scope and uncommented.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The D2 contracts audited function similarly to a traditional ETF. A user deposits assets into a Vault, where the funds are managed by the protocol and actively traded or invested using the D2 strategy over the course of an epoch. The strategy integrates with various DeFi protocols across multiple chains:

- **On Berachain:** Kodiak, Ooga Booga, Infrared, and Dolomite.
- **On Arbitrum One, Base, and Ethereum:** 1Inch, Aave, Pendle, GMX V2, Dolomite, and Silo.

Once the epoch ends, the trader returns the funds, and the proceeds are distributed among the vault shareholders in proportion to their holdings.

8.1 Key Components

- `VaultV0`**:** An ad-hoc trading vault that allows trading on protocols that do not yet have written integrations.
- `VaultV3`**:** The primary vault, which interacts with the `Strategy` diamond contract. Users deposit assets, which are then held by the `Strategy`, where the trader employs D2 strategies.
- `Strategy`**:** A diamond-style contract with immutable facets, each representing an integration with an external protocol.

8.2 VaultV3 Flow

1. A vault is created with a funding period and epoch length.
2. Users join by depositing assets during the funding period.
3. The epoch starts, and the funds are moved from the vault for trading.
4. The trader uses different modules in the Strategy to trade, invest, and earn yield for users.
5. The epoch ends, and the trader exits all positions and returns funds to the vault.
6. Users can withdraw their share of the earned profits.

9 Attestation

Cyfrin attests that it performed a security review of the D2 finance code provided by D2.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-02-24-cyfrin-d2-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to D2 has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=44c2c4447d500a172c334fff8e8f8b58352f11db6393317cbde6361ddbc3c199
F=2025-02-24-cyfrin-d2-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.